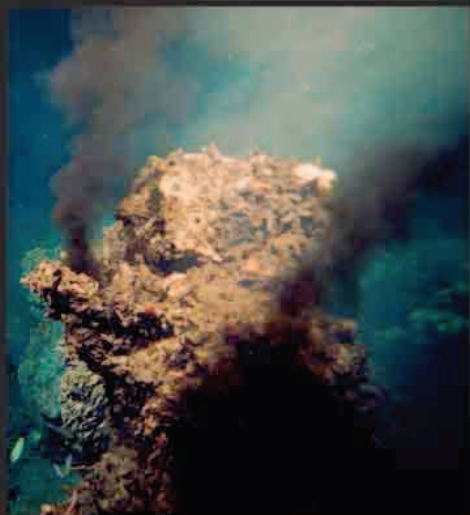


HOT CHEMICALS

The plumes of volcanically heated water that pour from black smokers are full of chemicals dissolved from hot rocks below the ocean floor. Microscopic archaea combine some of the chemicals with oxygen, and this releases the energy the microbes need to make sugar. These archaea use the sugar to build living cells, which provide food for other marine life.



▲ FOOD FACTORY

Thick mats of pale archaea smother the warm rock around an erupting black smoker.



HOT TAIL

The superheated water that erupts from volcanic vents is a source of life-supporting energy, but its high temperature can be deadly. Nevertheless, some animals can live amazingly close to it. This Pompeii worm lives with its head in a burrow where the water is around 68°F (20°C) and bathes its tail on the outside in temperatures of 158°F (70°C). This great contrast in temperatures would kill most animals.

Life on black smokers

The black smokers that boil up from mid-ocean ridges are the deep-sea equivalents of coral reefs—hotspots of teeming life amid the barren wastes of the deep ocean. Some extraordinary animals survive here because they have a source of food that does not rely on the energy of sunlight. Instead, the ecosystem is based on chemical energy from the black smokers themselves. Microbes called archaea use this energy to grow and multiply, and in turn feed colonies of animals including shrimp, clams, and giant worms.

SWARMING SHRIMP

The archaea living around the black smokers are grazed by swarms of animals that vary according to where they live. These white shrimp live on the Mid-Atlantic Ridge, and similar white crabs live around black smokers in the Pacific Ocean. The shrimp and crabs appear to be blind, but some have eyelike organs that may help them locate their food.



MUSSELS AND CLAMS

While some animals eat the archaea microbes living on the rocks, others feed on colonies of microbes living inside their bodies. These animals include giant mussels and clams that live around the vents, sucking chemically rich water into their shells to supply the microbes that live on their gills.

◀ VOLCANO MUSSELS

These clusters of giant mussels are living on a vent near the submerged Eifuku volcano in the western Pacific.